Global Blood Pressure Risk Intelligence Dashboard

Advanced SQL & Power BI Analysis — A comprehensive data analytics case study transforming raw health data into actionable global insights

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DATA ANALYTICS & BI

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Problem Statement

Identify High-Risk Populations

Analyze global blood pressure data to pinpoint which demographics and geographies carry the greatest cardiovascular risk burden.

Uncover Cross-Dimensional Patterns

Understand how blood pressure trends vary across countries and age groups to reveal hidden structural disparities.

Detect Data Inconsistencies

Surface discrepancies between individual patient-level data and aggregated country-level summaries that may mask true risk levels.

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Dataset Overview

Two complementary datasets were joined to enable both micro and macro-level analysis, with key columns spanning demographics, vitals, and regional identifiers.

Patient-Level Dataset

Individual records including systolic BP, diastolic BP, age, and country of origin — the granular foundation of all risk analysis.

Country-Level Summary Dataset

Pre-aggregated national averages for BP metrics, used to benchmark individual readings and detect reporting deviations.

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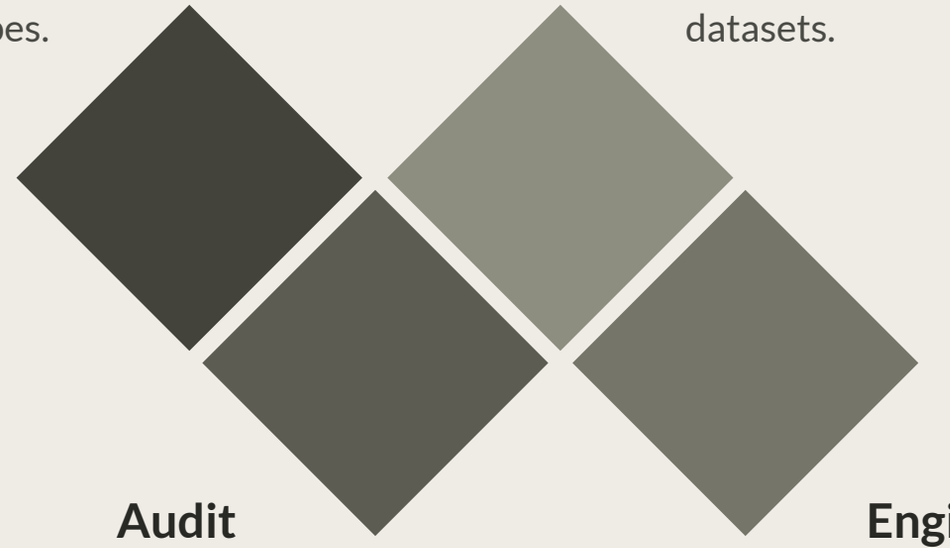
Data Preparation & Cleaning

Standardize

Aligned column formats and types.

Validate

Ensured join integrity across datasets.



Audit

Checked for nulls and inconsistencies.

Engineer

Created derived features for analysis.

A rigorous preparation phase ensured analytical accuracy. Every column was inspected, every data type validated, and both datasets were harmonized before any analysis was performed — eliminating noise and building a reliable analytical foundation.

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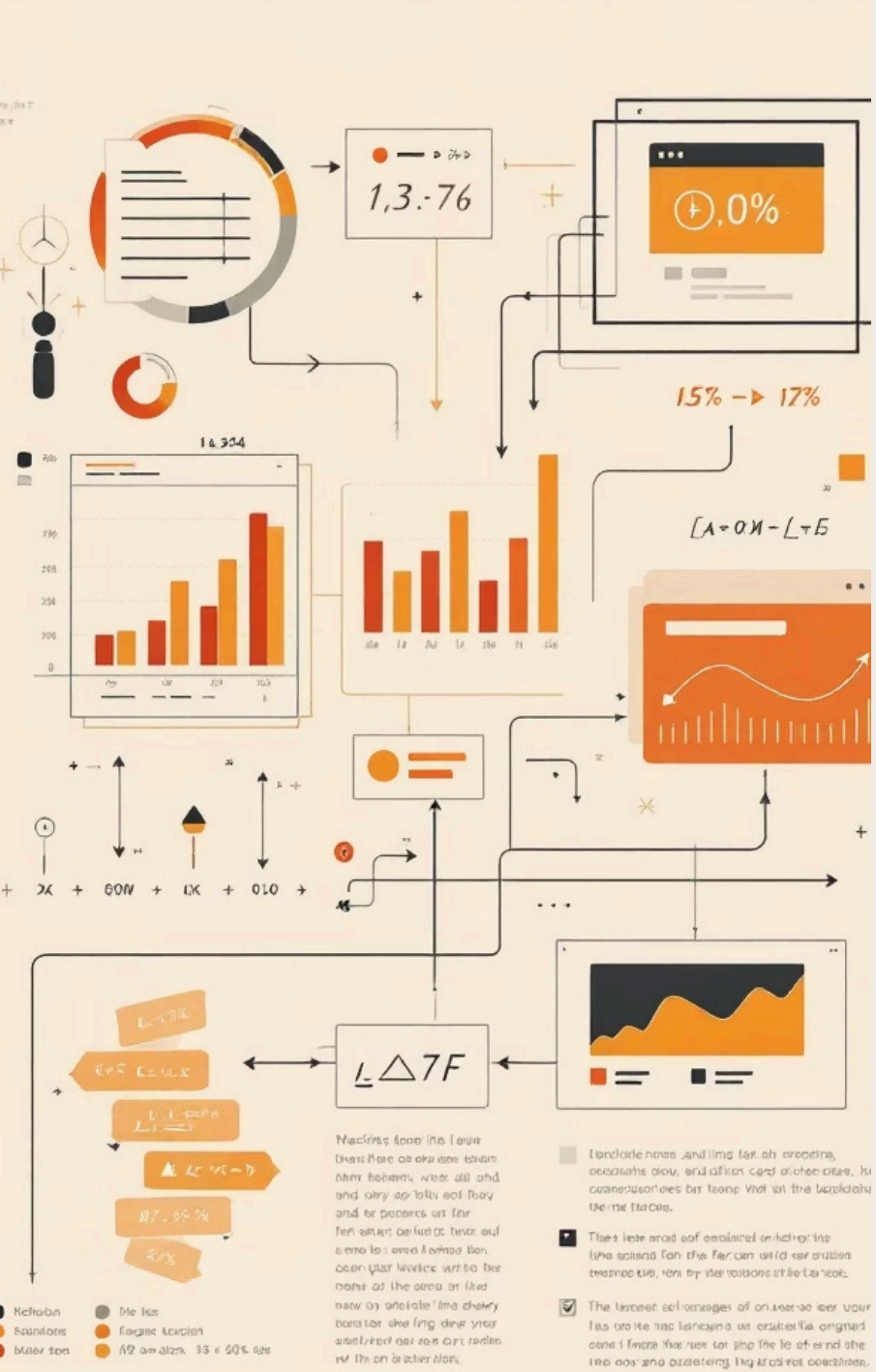

Feature Engineering — Key Innovation

Age Groups

Segmented patients into clinically meaningful age bands to enable cohort-level analysis and trend comparison.

Risk Category

Classified patients as Normal, Elevated, or High-Risk using established medical BP thresholds for actionable segmentation.

Composite Risk Score

Built a weighted scoring model combining systolic BP, diastolic BP, and age — revealing hidden high-risk individuals beyond standard thresholds.

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SQL Analysis — Advanced Techniques

1

JOINs

Combined patient and country datasets for integrated cross-dimensional analysis.

2

CASE WHEN

Applied conditional logic to dynamically classify risk categories across the full dataset.

3

Window Functions

Used RANK and NTILE to perform percentile analysis and isolate the top 1% highest-risk patients.

4

Deviation Analysis

Compared actual patient readings against reported country averages to surface hidden discrepancies.

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Dashboard Development — Power BI

PAGE 1: EXECUTIVE OVERVIEW

KPIs at a glance, country-level BP trend maps, and risk category distribution — designed for senior stakeholders who need rapid situational awareness.

PAGE 2: DEEP INSIGHTS

Advanced risk segmentation, age-group patterns, and composite score visualizations — built for analysts seeking granular exploration.

Dynamic **DAX measures** power all calculations. Slicers for **country**, **age group**, and **risk category** provide full interactive drill-down capability.

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Key Insights

Regional Clustering

Certain countries consistently exhibit elevated BP levels, with high-risk patients clustering in specific geographic regions rather than being evenly distributed.

Hidden High-Risk Individuals

Composite risk scoring surfaces patients who fall below standard thresholds but carry significant cumulative cardiovascular risk.

Age-BP Correlation

Age shows a strong positive correlation with blood pressure, validating the need for age-stratified intervention strategies.

Reporting Discrepancies

Several countries show measurable gaps between officially reported BP averages and actual patient-level data, raising data quality flags.

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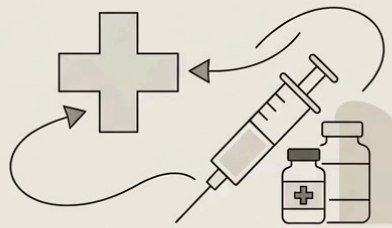
Healthcare & Business Impact

Early Risk Detection
(identify high-risk populations before crisis)



Targeted Interventions
(support precision healthcare resource allocation)

Policy Intelligence

Policy Intelligence
(equip policymakers with regional risk maps)

Data-Driven Decisions

Data-Driven Decisions
(demonstrate analytics as a healthcare decision engine)

From Data to Decisions

This dashboard transforms raw health metrics into a **strategic decision-support tool** — enabling healthcare organizations and policymakers to move from reactive treatment to proactive, data-driven population health management.

The analytical framework is designed to scale — applicable to other chronic conditions and expanded datasets with minimal rework.

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Tools, Technologies & Conclusion

Advanced SQL

JOINs, CTEs, Window Functions, CASE WHEN, and deviation analysis for end-to-end data engineering and querying.

Power BI & DAX

Interactive 2-page dashboard with dynamic DAX measures, data modeling, and slicer-driven interactivity.

Data Storytelling

Transformed raw health data into actionable intelligence — delivering a scalable, insight-driven analytics solution ready for real-world deployment.

Project Outcome: Successfully delivered an end-to-end analytics pipeline — from raw data ingestion through SQL engineering to interactive Power BI visualization — producing measurable, clinically relevant insights at global scale.

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